

YourTurn: Manual Quantile Regression (Linear Model)

Understanding the Check Function in Excel

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Purpose

The purpose of this worksheet is to demonstrate **how quantile regression is estimated**.

Before using the `rq()` function in R, you will manually fit a quantile regression model by changing the regression coefficients and observing how the **total check loss** changes.

Rather than treating quantile regression as a “black box”, this exercise shows the optimisation process step by step.

The Data

The spreadsheet uses the **Engel** dataset from the **quantreg** package.

```
library(quantreg)
```

Loading required package: SparseM

```
data(engel)
```

```
head(engel)
```

```
      income  foodexp
1 420.1577 255.8394
2 541.4117 310.9587
3 901.1575 485.6800
4 639.0802 402.9974
5 750.8756 495.5608
6 945.7989 633.7978
```

The Engel dataset contains **235 Belgian working-class households** observed in 1857 by Ernst Engel.

For each household we observe

- **income** - annual household income (Belgian francs)
- **foodexp** - annual household food expenditure (Belgian francs)

Our aim is to estimate the conditional relationship between food expenditure and income.

The Spreadsheet

The spreadsheet implements the quantile regression objective function directly.

Your task is to change the regression coefficients until the **Total Check Loss** is as small as possible.

Column A: Income

Column **A** contains the predictor variable

$$x_i = \text{income.}$$

These values come directly from the **engel** dataset.

Column B: Food Expenditure

Column **B** contains the observed response

$$y_i = \text{food expenditure.}$$

These are the values we wish to model.

Columns C and D: Regression Coefficients

These cells contain the coefficients of the regression model

- β_0 (intercept)
- β_1 (slope)

Together they define the fitted regression equation

$$\hat{y}_i = \beta_0 + \beta_1 x_i.$$

Try changing these coefficients and observe how the fitted regression line changes.

Column E: Predicted Values

Column **E** calculates the fitted values

$$\hat{y}_i = \beta_0 + \beta_1 x_i.$$

These are the predicted food expenditures given your current regression coefficients.

Column F: SIGN(u)

Column **F** calculates

```
=SIGN(B2-E2)
```

where

$$u_i = y_i - \hat{y}_i$$

is the residual.

The SIGN function determines on which side of the fitted line each observation lies.

SIGN(u)	Interpretation
1	Observation lies above the fitted line ($u > 0$)
0	Observation lies on the fitted line
-1	Observation lies below the fitted line ($u < 0$)

This column is an intermediate step used to construct the indicator function.

Cell G2: Quantile Level

Cell **G2** contains the quantile level

τ .

Typical values are

- $\tau = 0.25$
- $\tau = 0.50$
- $\tau = 0.75$

Changing τ changes the relative penalty assigned to positive and negative residuals.

Column H: Indicator Function

Column **H** calculates

```
=IF(F2<0,1,0)
```

which implements the indicator function

$$1(u < 0).$$

The indicator function acts as a **switch**.

If the residual is negative,

$$u < 0,$$

the indicator returns

$$1.$$

Otherwise it returns

$$0.$$

Its only purpose is to determine which weight should be applied inside the check function.

Column I: The Check Function

Column **I** calculates

```
= (B2-E2)*($G$2-H2)
```

This is the Excel implementation of the check function

$$\rho_{\tau}(u) = u(\tau - 1(u < 0)).$$

The indicator function $1(u < 0)$ acts as a **switch**. It checks whether the residual is negative, that is, whether an observation lies below the fitted line. If $u < 0$, the indicator equals 1; otherwise it equals 0. This allows the check function to apply different penalties to observations on opposite sides of the fitted line without requiring separate formulas.

The parameter τ determines the size of those penalties. Positive residuals (observations above the fitted line) are weighted by τ , whereas negative residuals (observations below the fitted line) are weighted by $1 - \tau$. By changing these relative weights, quantile regression shifts the fitted line towards different parts of the conditional distribution. For example, when $\tau = 0.25$, negative residuals receive a larger penalty than positive residuals, pushing the fitted line downward towards the lower tail of the distribution. Conversely, when $\tau = 0.75$, positive residuals are penalised more heavily, moving the fitted line upward towards the upper tail.

Each row therefore contributes **one check loss** to the total objective function.

Column J: Total Check Loss

Column J computes

```
=SUM(I2:I236)
```

which is

$$\sum_{i=1}^{235} \rho_{\tau}(u_i).$$

This quantity is called the **Total Check Loss**.

It is the objective function that quantile regression minimises.

As you change the regression coefficients,

- the fitted line moves,
- the residuals change,
- the check losses change,
- and the Total Check Loss updates automatically.

Your goal is to make this value **as small as possible**.

Columns K-R

The remaining columns contain helper calculations used for plotting the regression lines and charts.

They are **not part of the optimisation** and can safely be ignored during this exercise.

Your Task

1. Choose a value of τ .
 2. Change the regression coefficients
 - β_0
 - β_1
 3. Observe how the fitted regression line changes.
 4. Watch the **Total Check Loss** update.
 5. Continue adjusting the coefficients until the Total Check Loss is as small as possible.
 6. Record the coefficients that minimise the objective function.
-

Comparing with R

Once you have found the minimum manually, compare your solution with the estimates obtained from

```
library(quantreg)

rq(
  foodexp ~ income,
  data = engel,
  tau = 0.50
)
```

The `rq()` function performs **exactly the same optimisation** that you have carried out manually.

The only difference is that `rq()` uses an efficient optimisation algorithm to find the minimum instantly, whereas you have located it by trial and error.

Take-Home Message

This spreadsheet implements quantile regression exactly as defined mathematically.

For each observation it

1. computes the fitted value,
2. determines whether the observation lies above or below the fitted line,
3. evaluates the check loss,
4. sums the losses over all observations.

Quantile regression then chooses the regression coefficients that minimise this total check loss.

The `rq()` function simply automates this optimisation.